

Paper on template assisted DWCNT growth

Samples in Solution

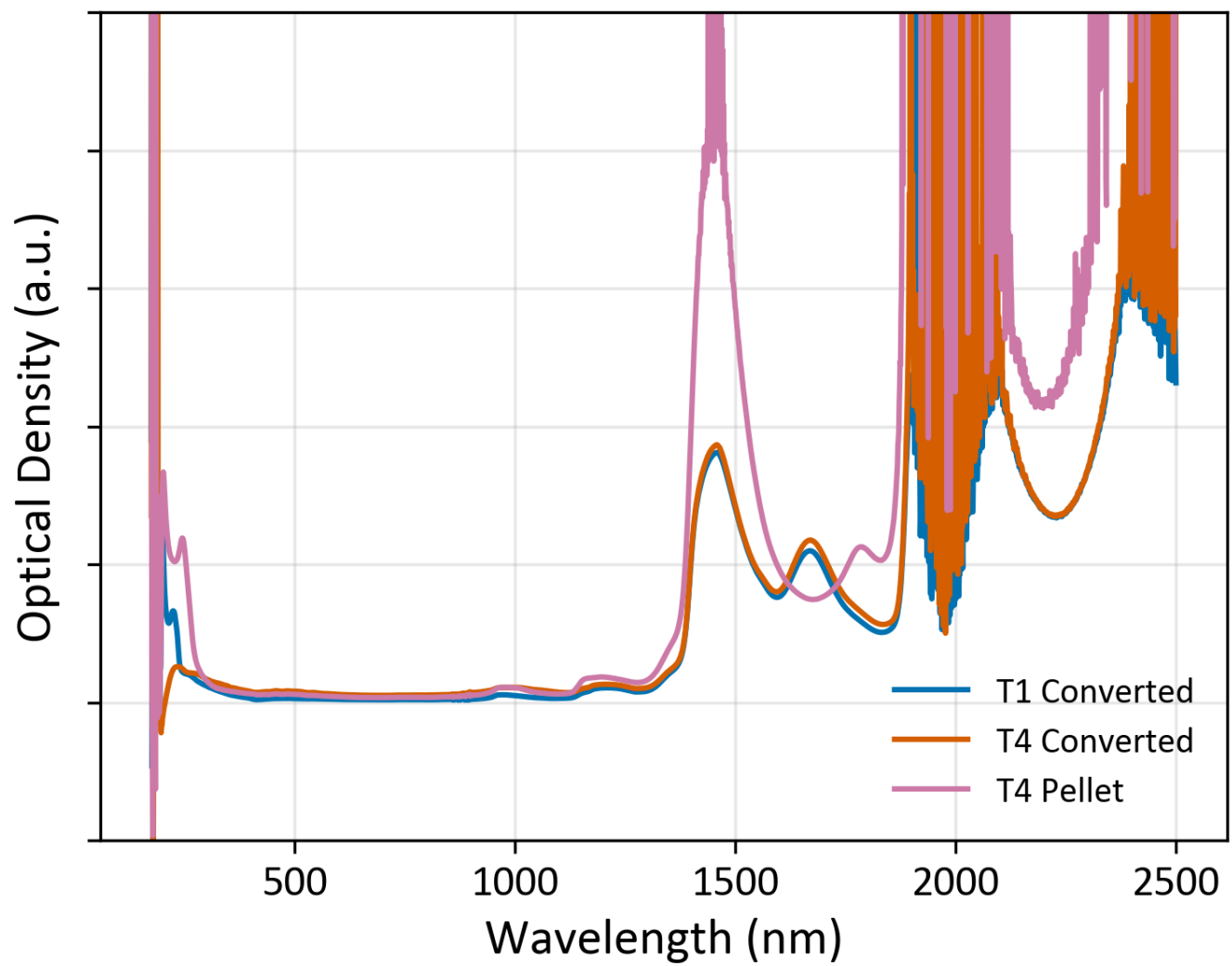
We received:

- sT1 converted dispersion
- sT4 converted dispersion
- sT4 converted dispersion - Pellet after PFO wrapping and DGU

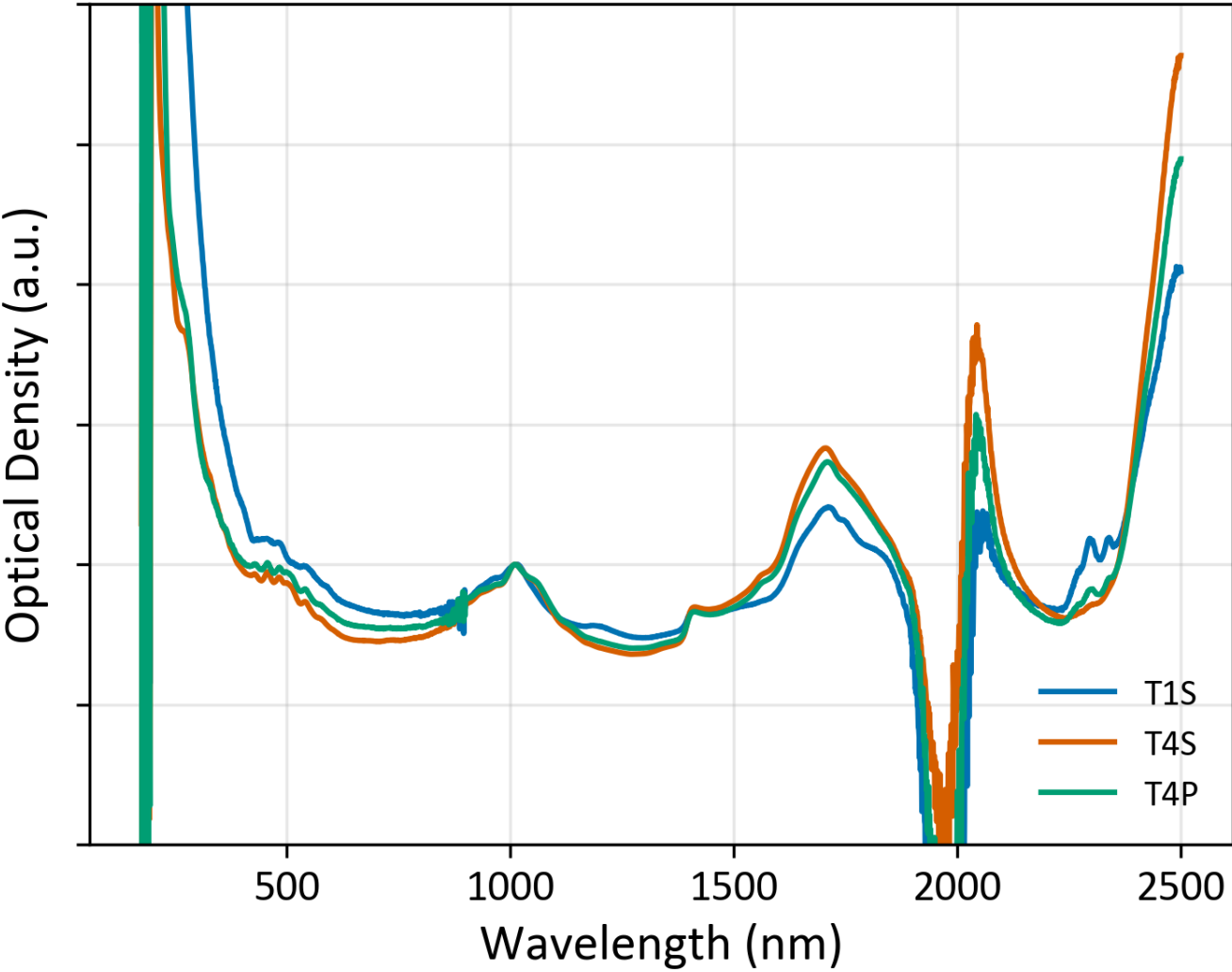


Absorption

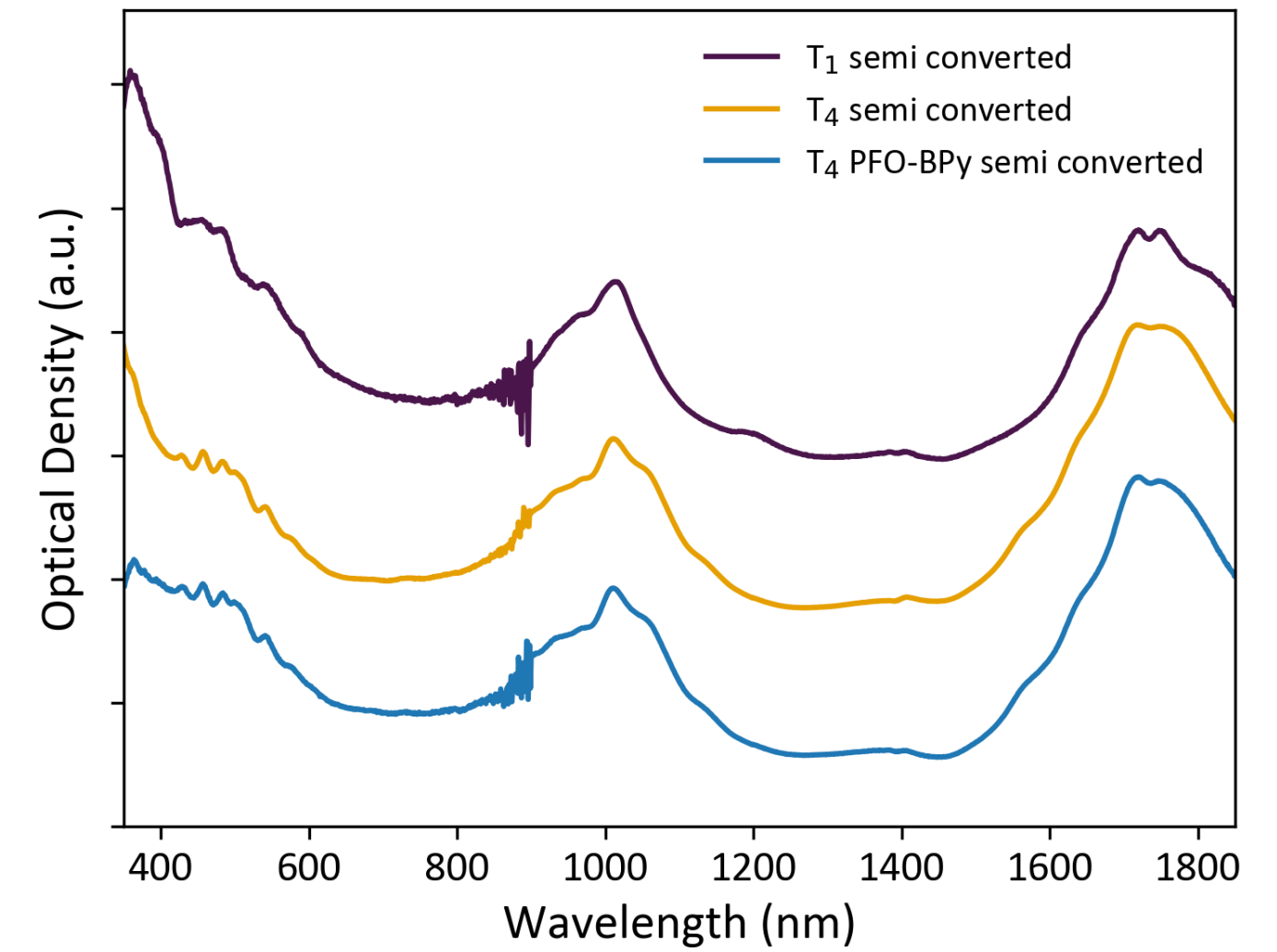
- Too much water content:



- After this we did a dialysis to remove water content. Once in D2O we re-measured absorption:

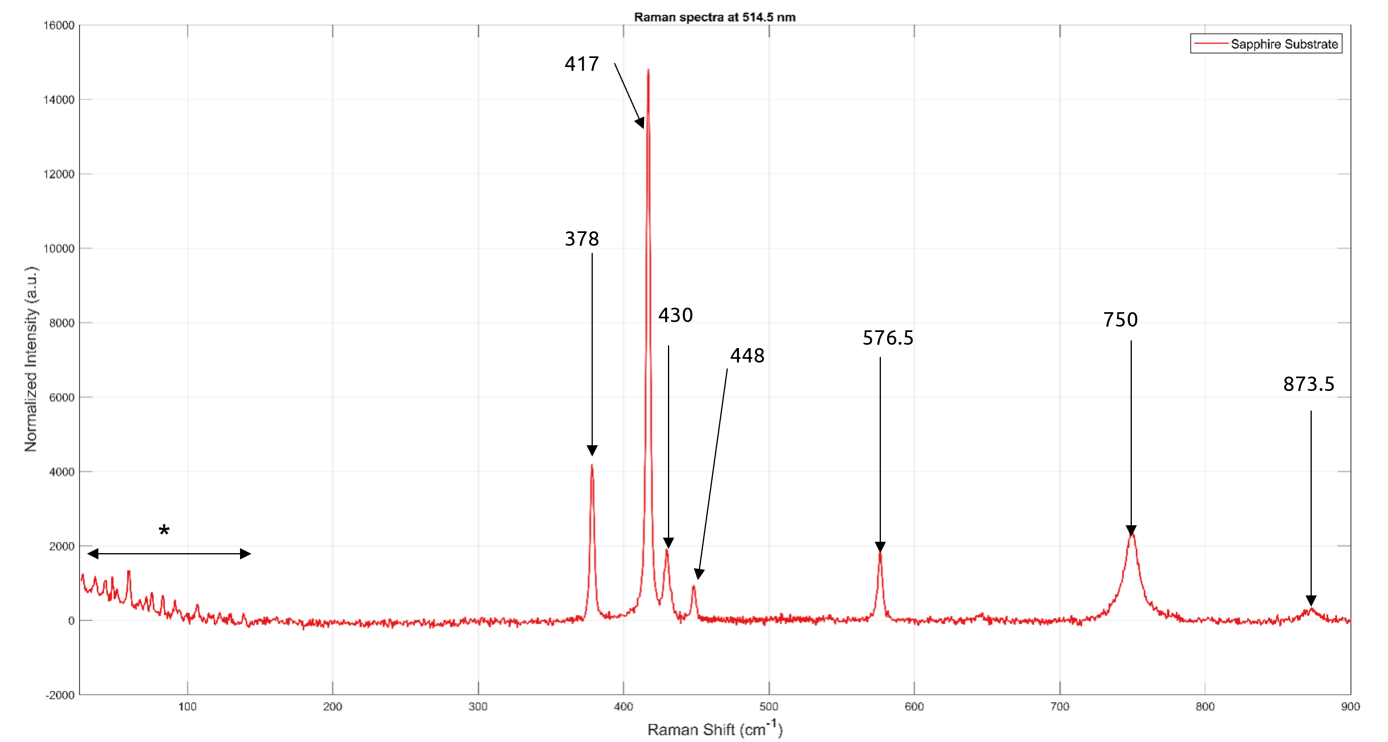


- After substracting water baseline + offset:



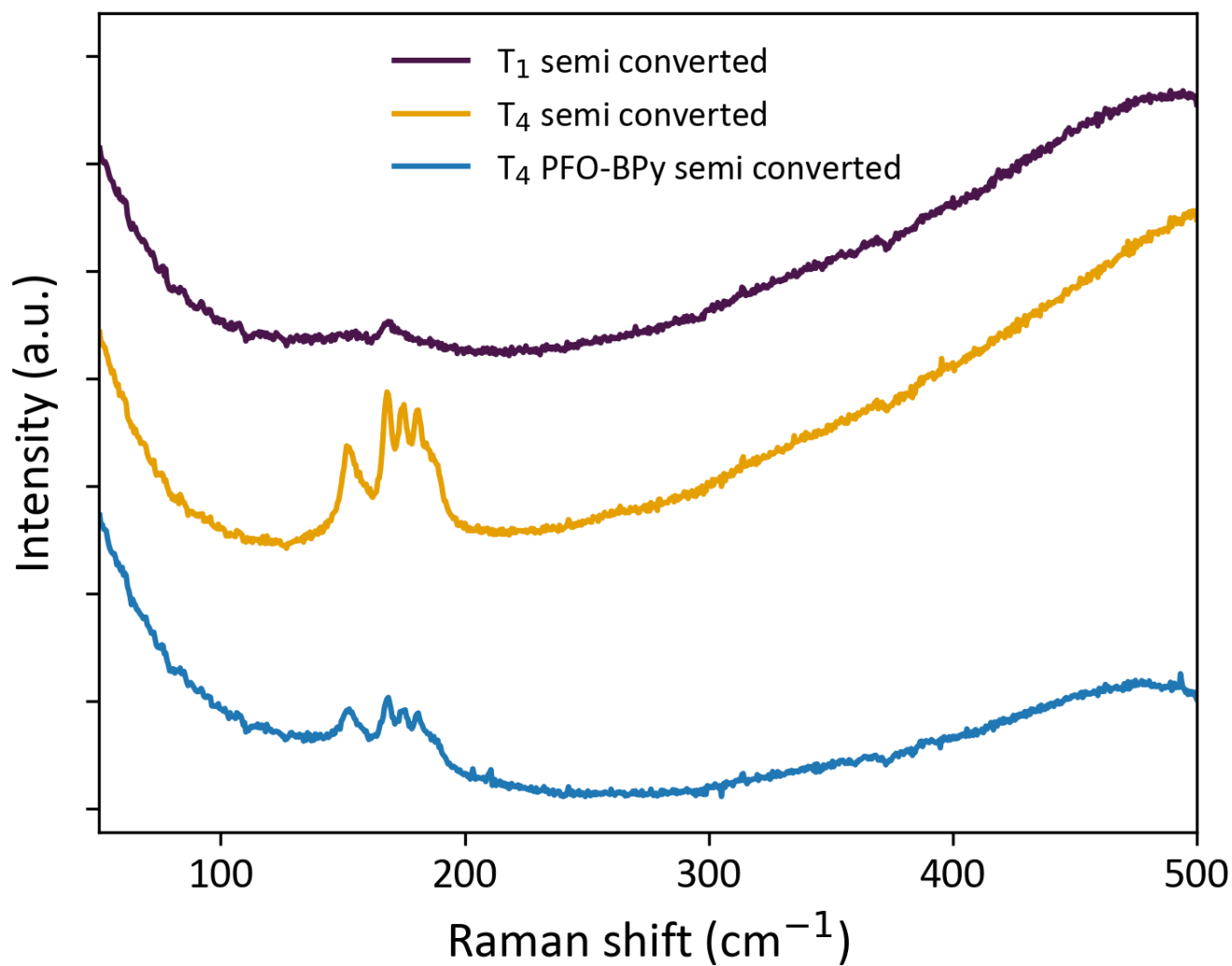
Raman

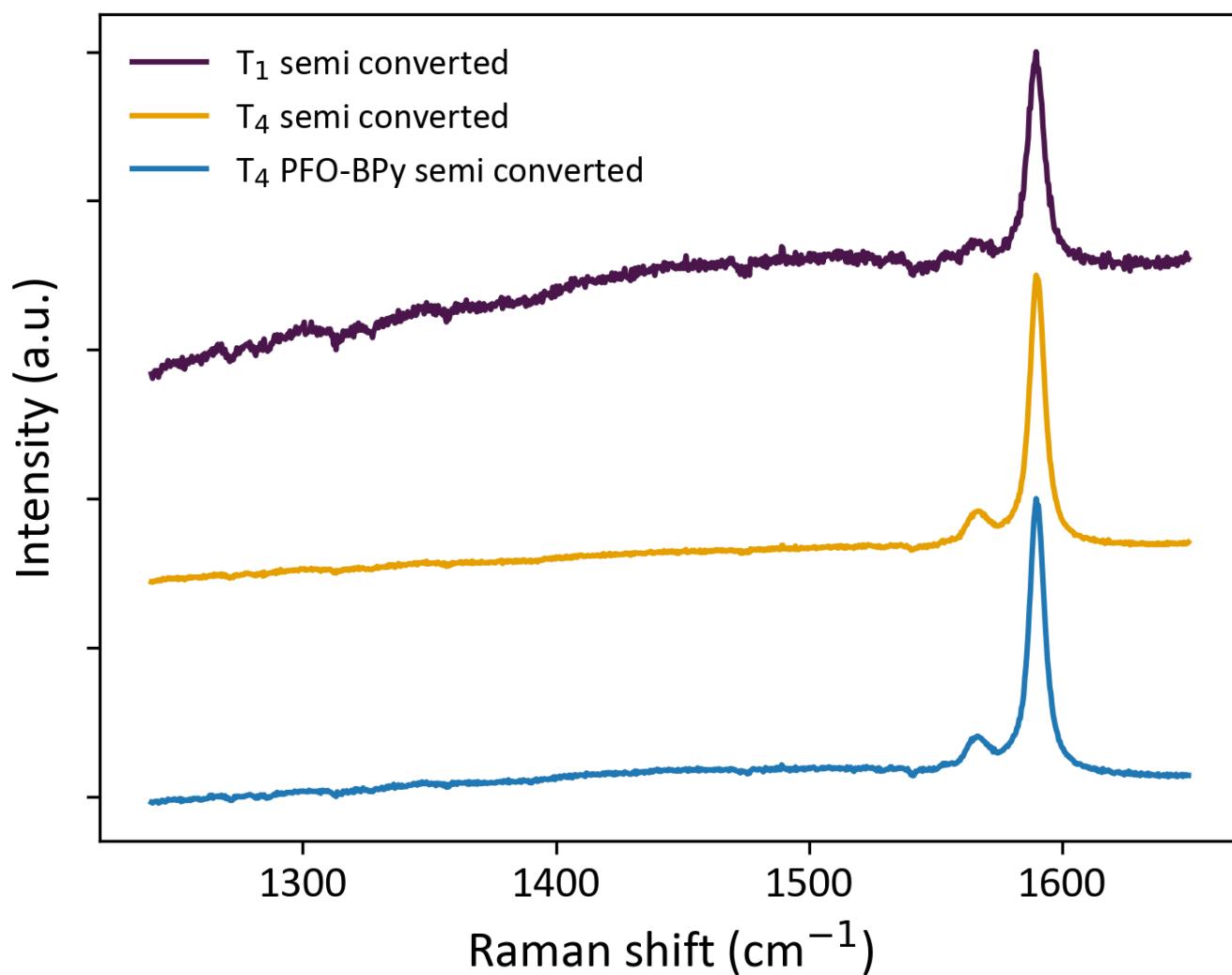
- For reference here is the spectrum of the substrate:



Raman@514nm

- Raman on the same dispersion samples as shown above. We observed a fluorescence background when measuring Raman (it was there even before the dialysis step)
- The G band does not show the splitting that was observed in the other film samples after conversion.





Initial films for testing

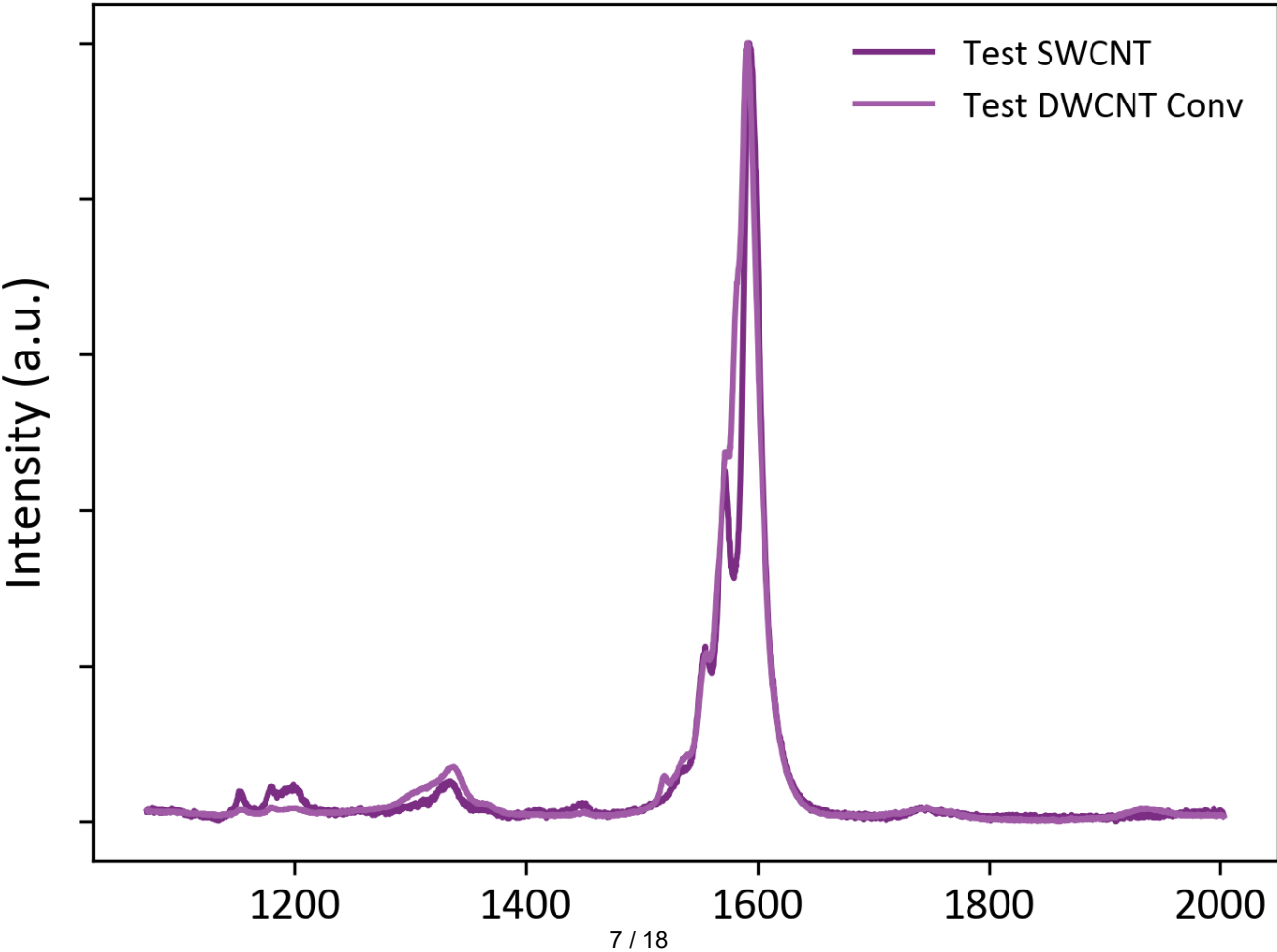
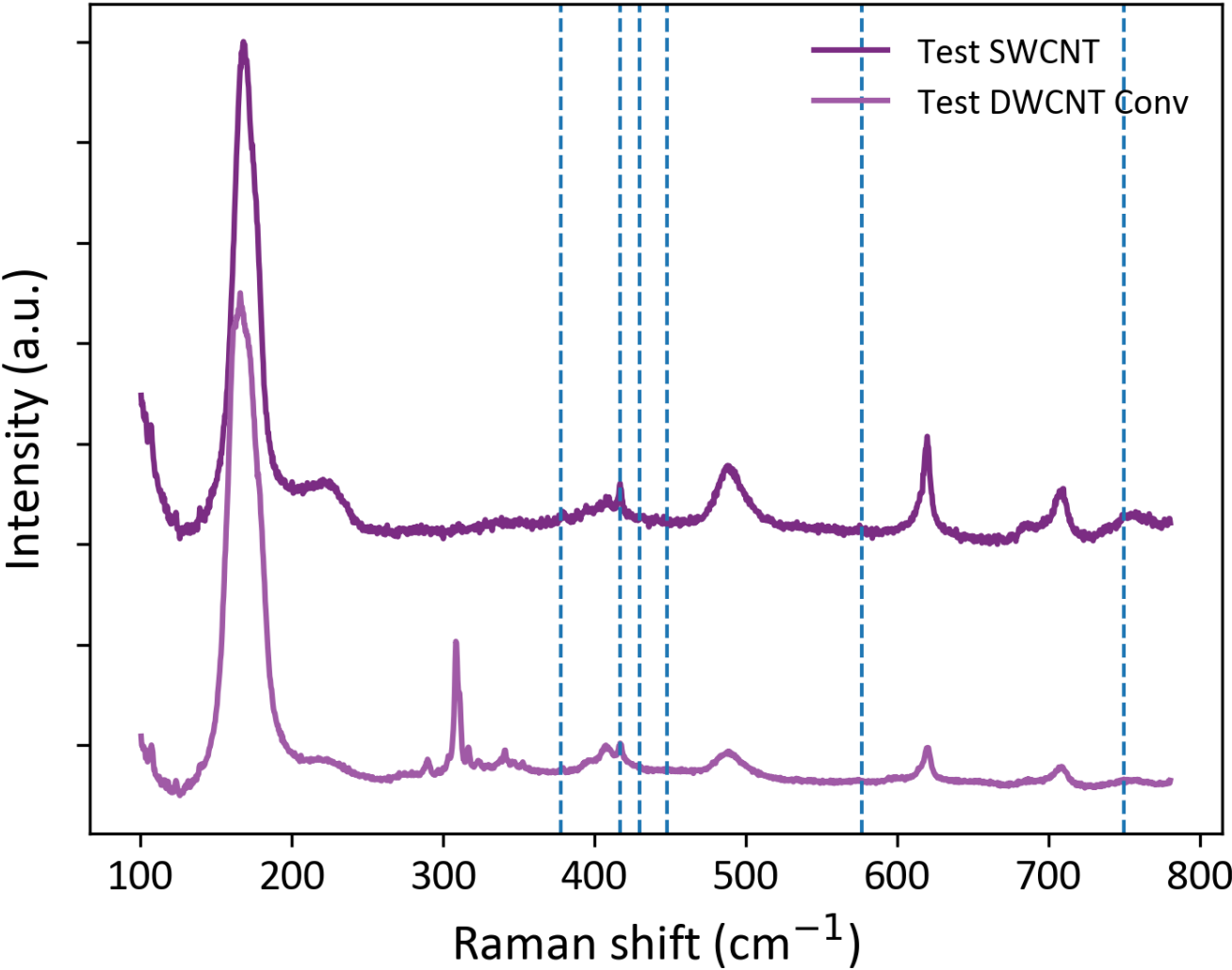
We received:

- film before conversion
- film after conversion



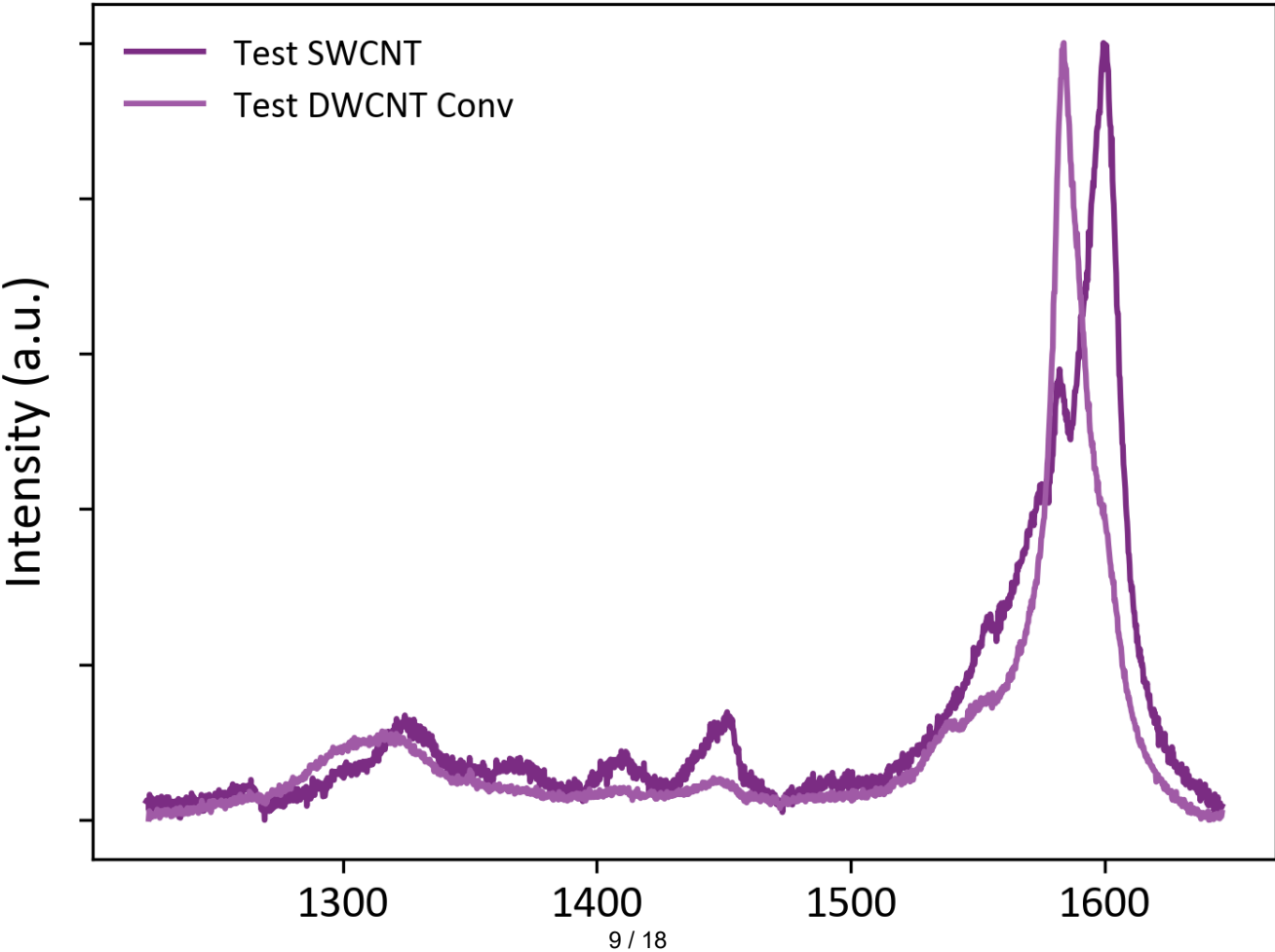
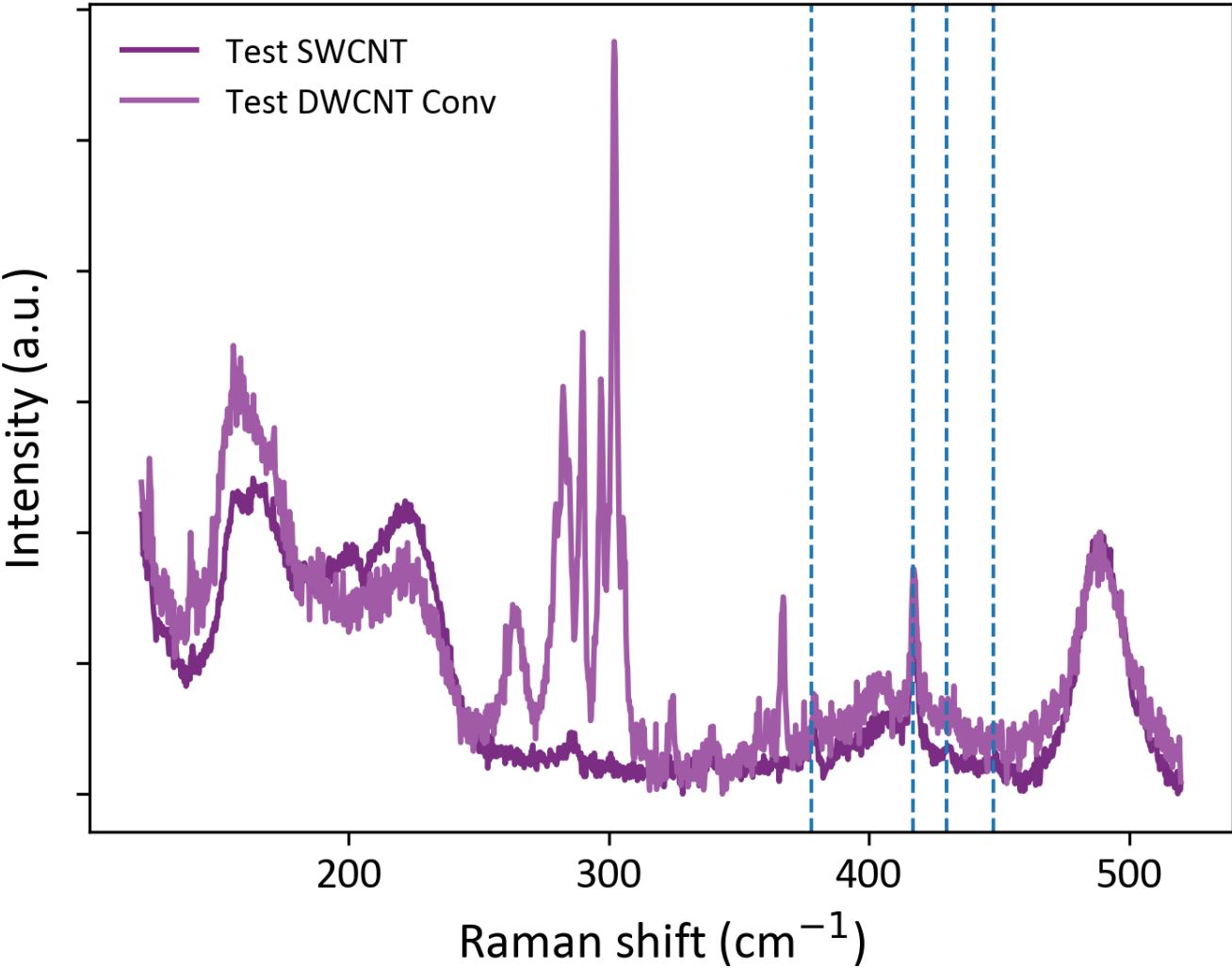
In the Raman measurements, the dotted lines correspond to the peak positions identified on the sapphire substrate.

Raman@570nm



Raman shift (cm^{-1})

Raman@650nm

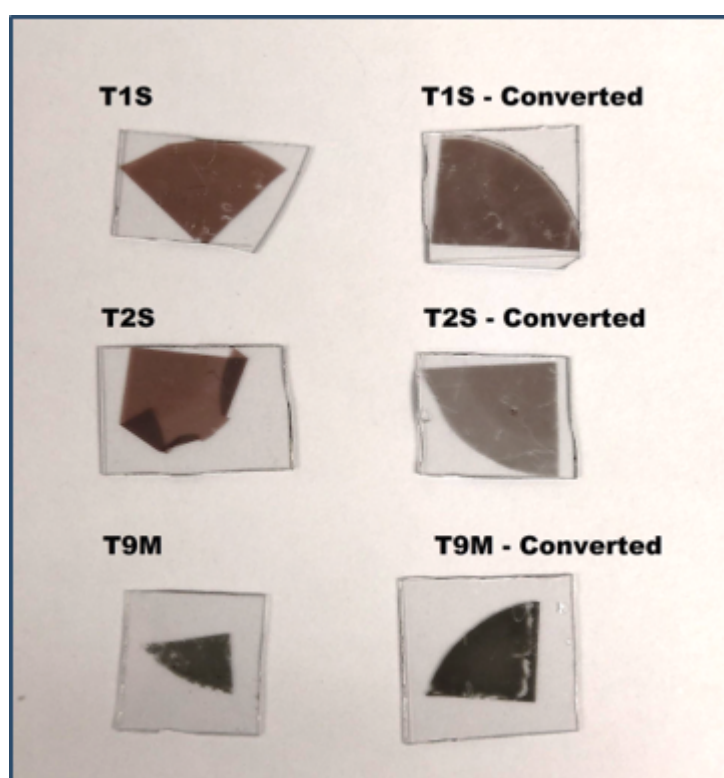


Raman shift (cm^{-1})

Films before and after conversion

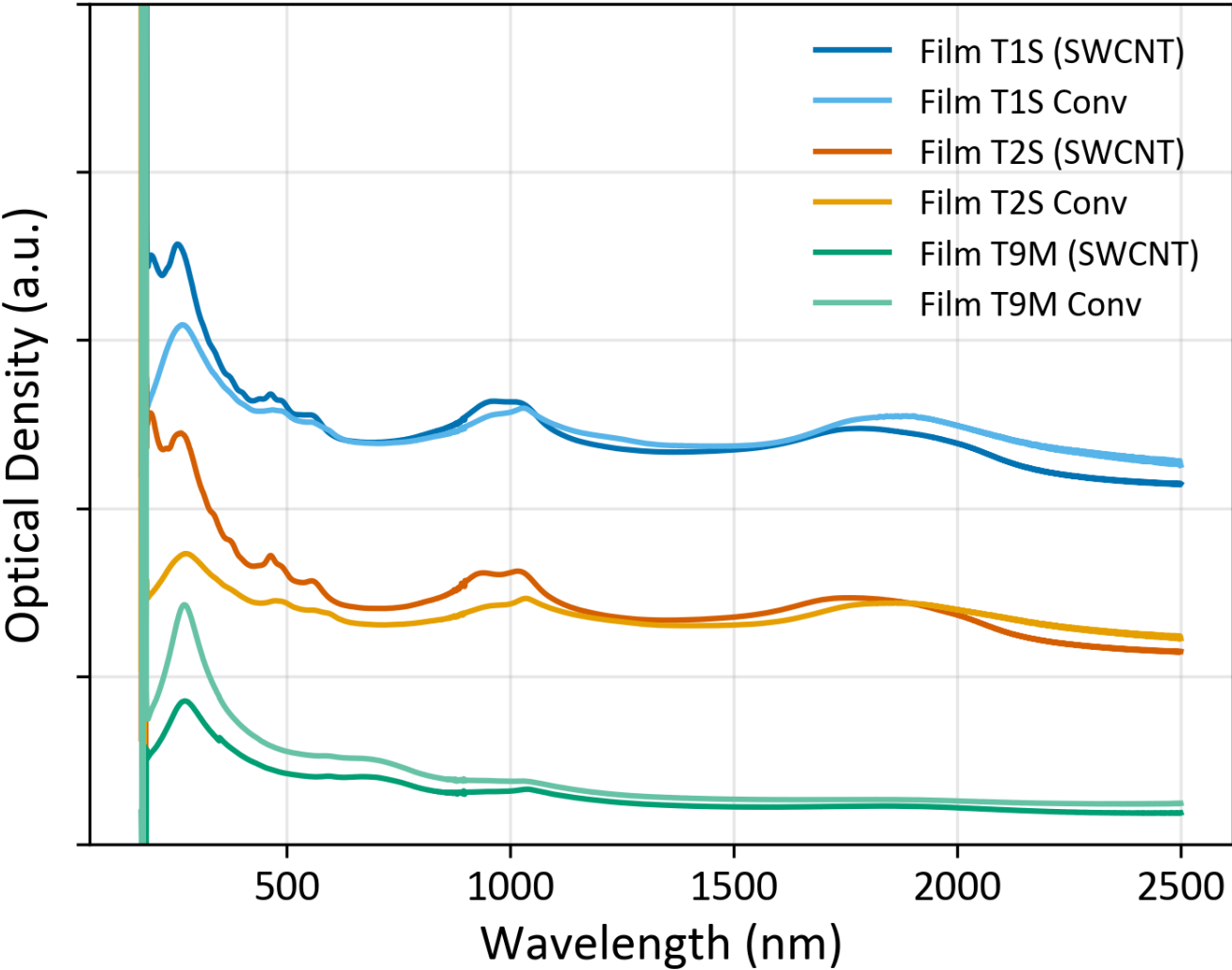
We received:

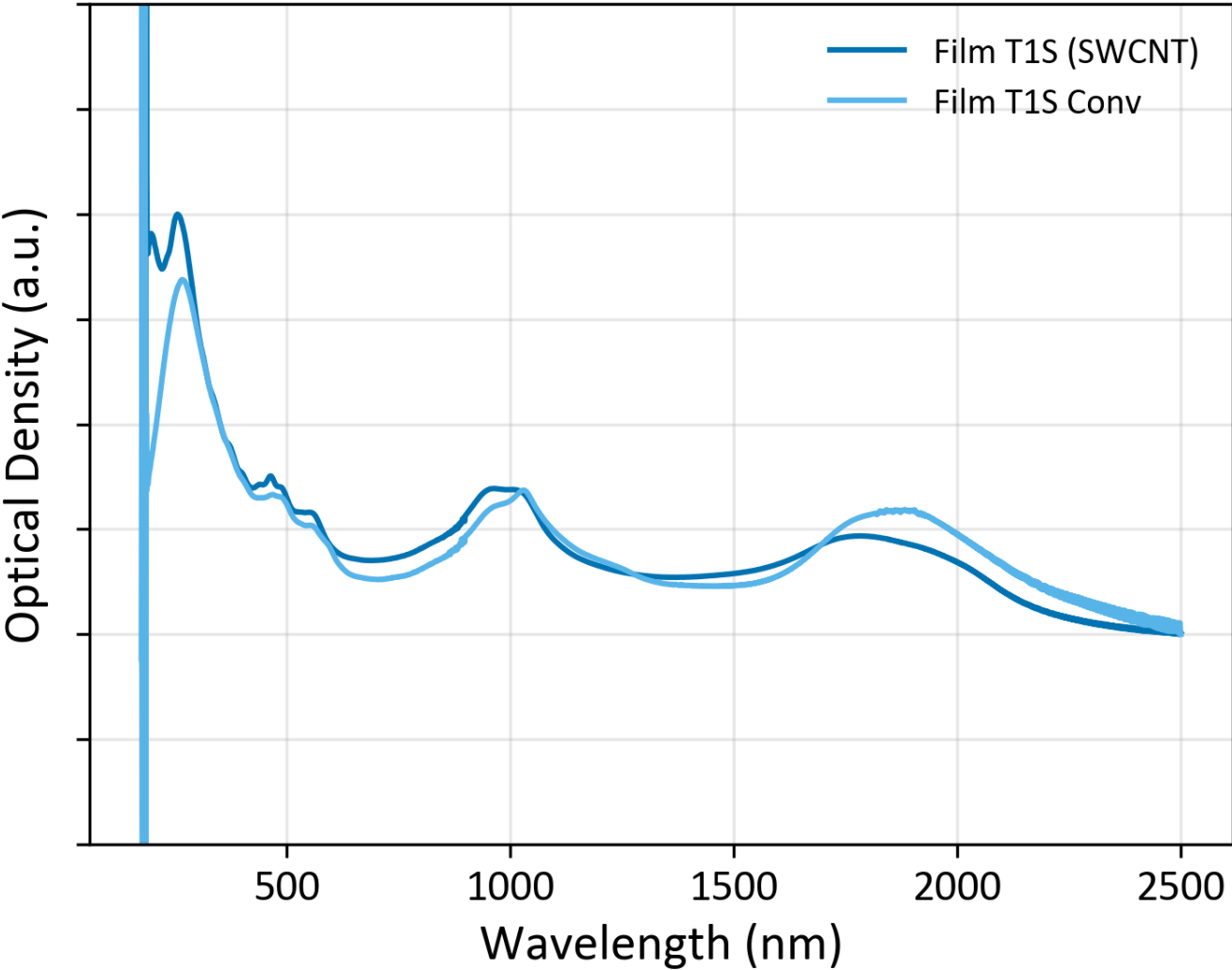
- sT1 film before conversion
- sT1 converted film
- sT2 film before conversion
- sT2 converted film
- mT9 film before conversion
- mT9 converted film

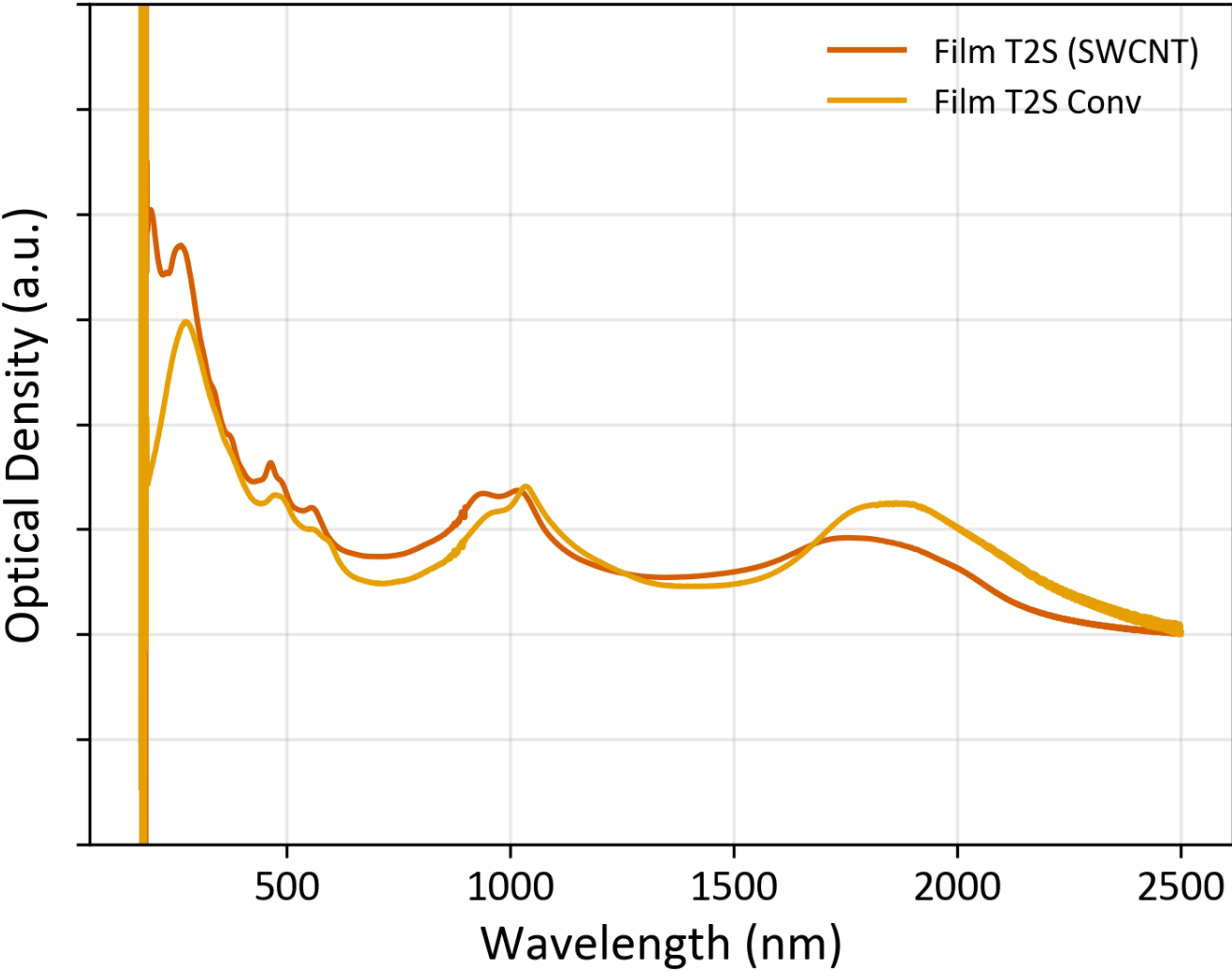


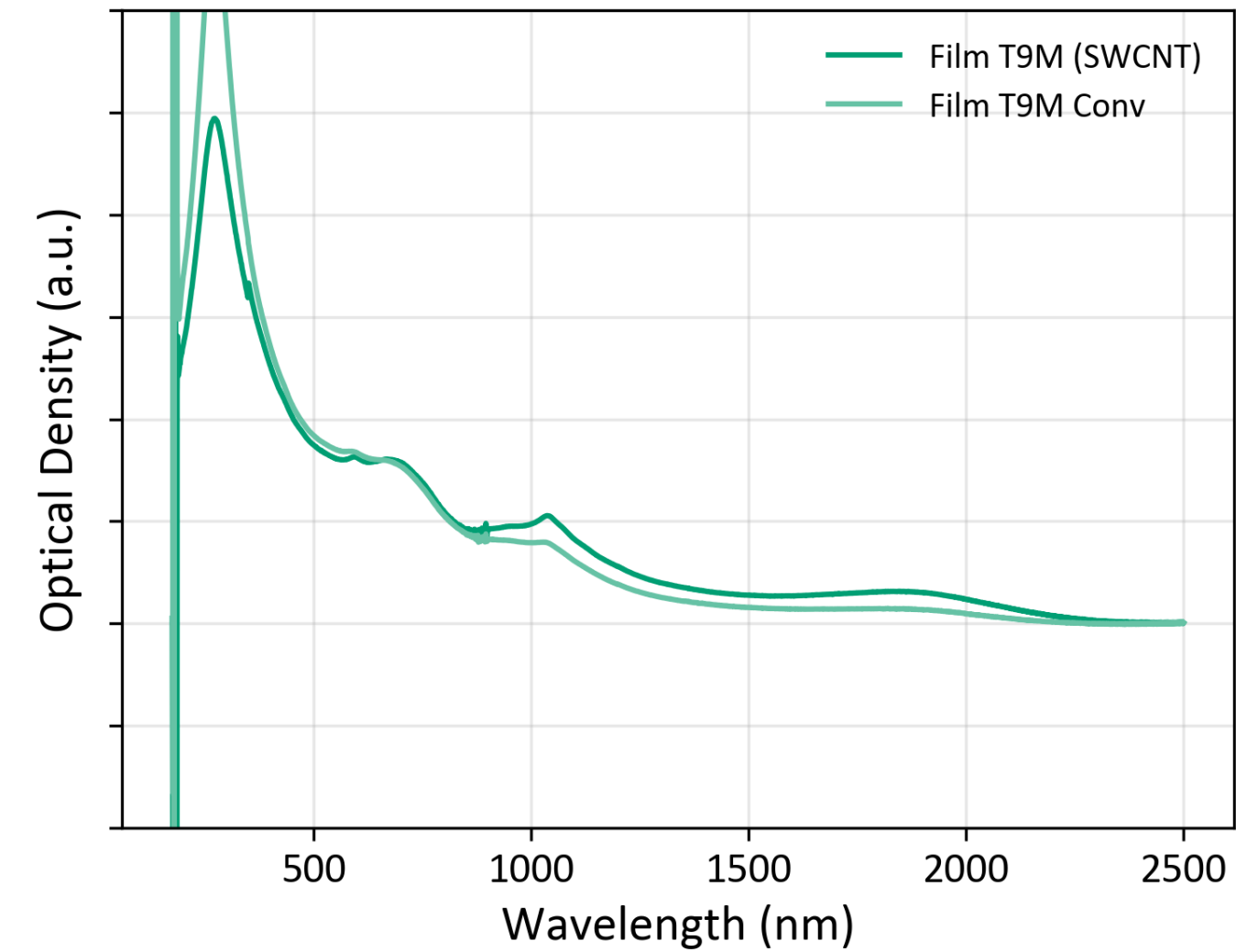
Absorption

More prominent metallic peaks after conversion. Results consistent with figure 2 of the manuscript.

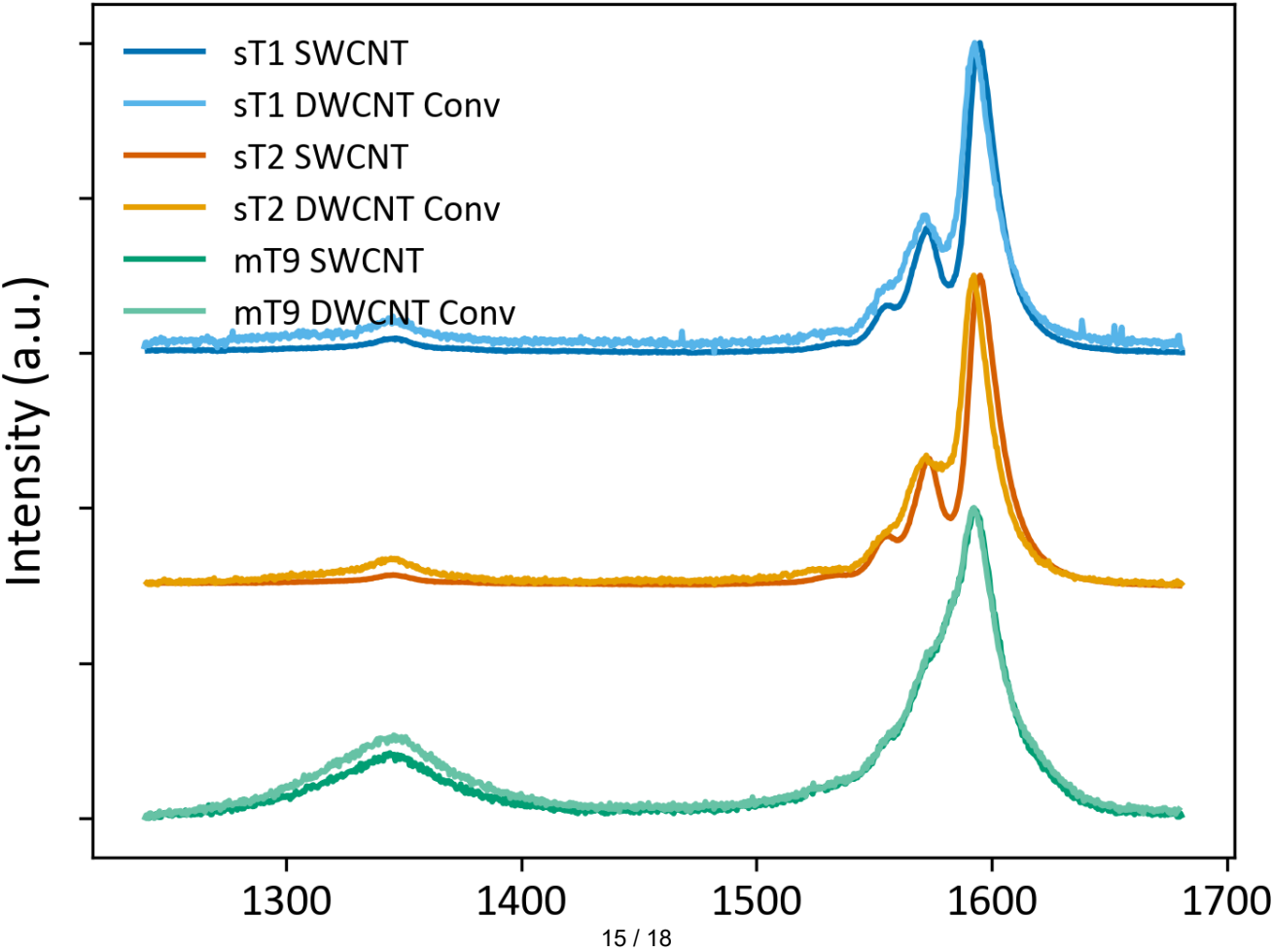
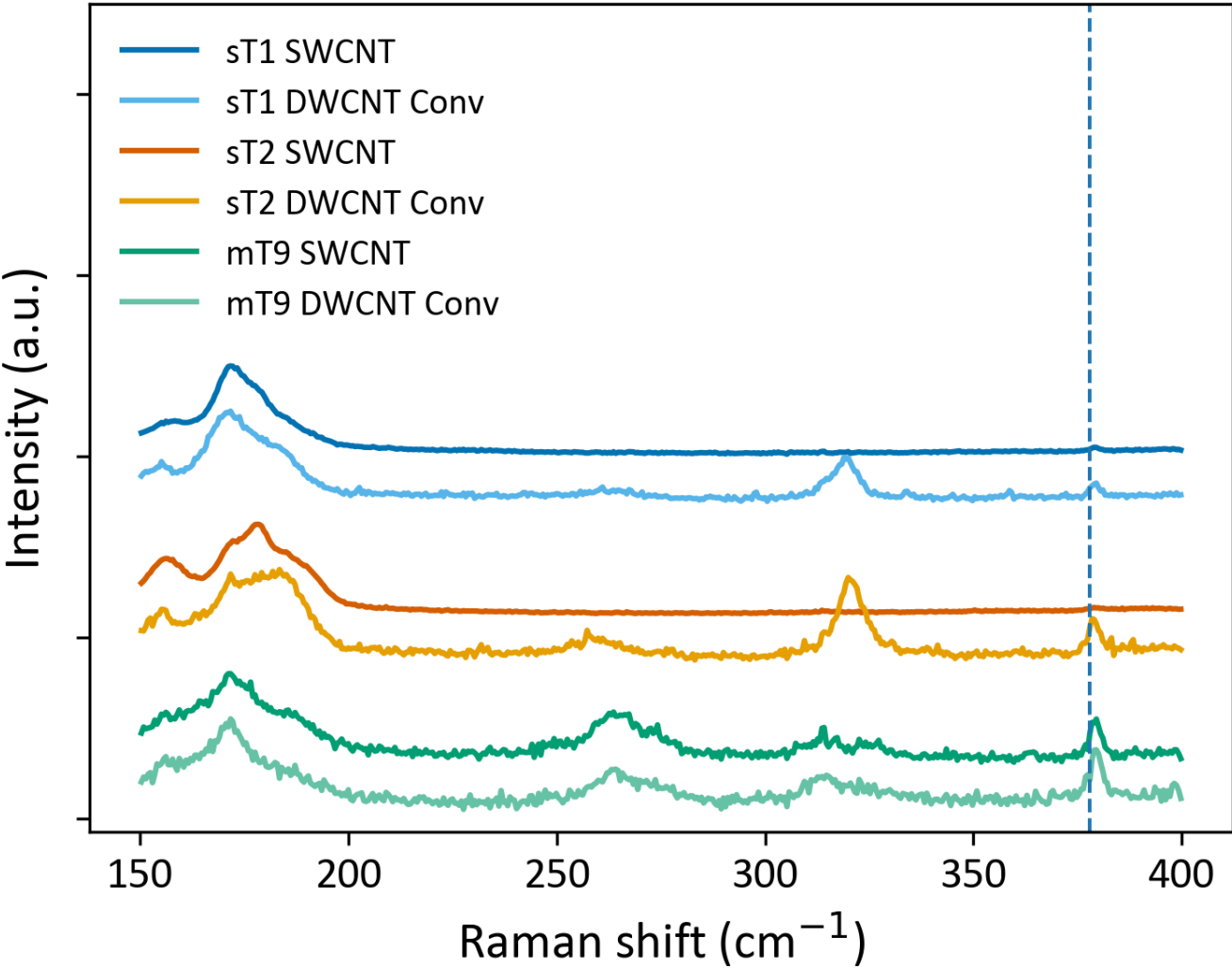






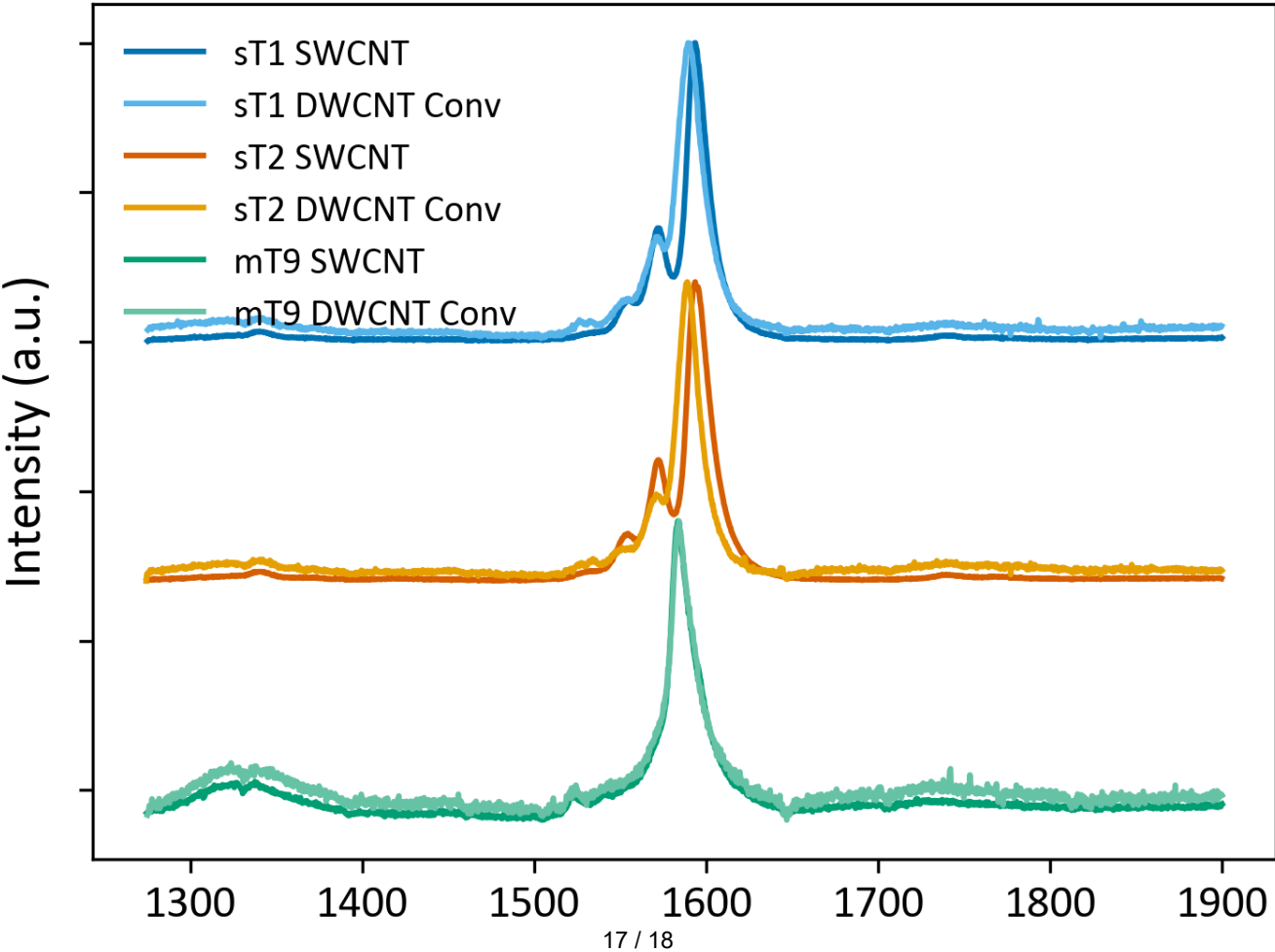
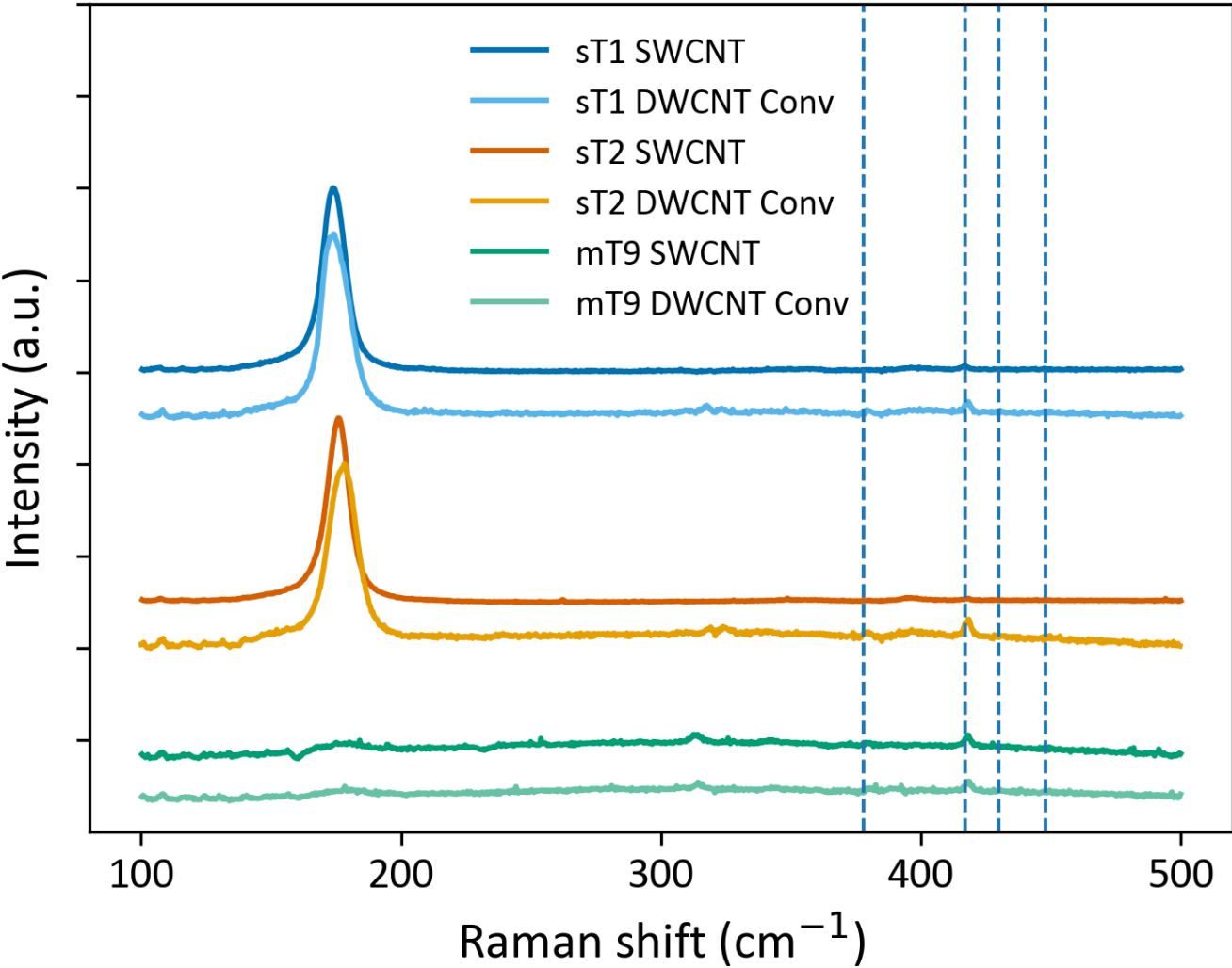


Raman@514nm



Raman shift (cm^{-1})

Raman@561nm



Raman shift (cm^{-1})